Hongxiang Wang

Hxjob1017@gmail.com | (314) 753-3414 | github.com/DanielWHX

EDUCATION

University of Illinois Urbana-Champaign
Master of Computer Science — Jan. 2027 - Dec. 2028

The Ohio State University
Bachelor of Art and Science in Computer Science — Aug. 2021 - Apr. 2026

- **Relevant Coursework:** Java, Data Structures and Algorithms, Operating Systems, Web Applications, Database Systems, Computer Networks, Software Engineering, Information Security, Requirements Analysis, Python

EXPERIENCE

Lyntra LLC (Startup) — Remote
Full-stack Developer Intern — Nov. 2025 - Mar. 2026

*Built **Node.js/TypeScript backend services** for an AI scheduling platform integrating students' Calendar/Canvas coursework with an LLM-based planner, covering **3 core workflows: adaptive scheduling, task decomposition, and external-data ingestion**.*

- **Reduced estimated LLM inference cost by ~98%** by redesigning the Planning Agent's **model-selection and inference pipeline**, using structured prompting, schema-constrained outputs, and fallback execution to reliably decompose high-level assignments into actionable subtasks.
- Developed the core **adaptive scheduling workflow** that converted homework and project into daily task plans, adding conflict detection and automatic rescheduling, which reduced **75%+** manual adjustments for users
- Built an **LLM-based Agent with a Think–Execute loop**, integrating Tool Calling and pgvector-backed RAG to retrieve the **top-3 relevant chunks in sub-500ms** and support grounded multi-step workflows.

Petro-CyberWorks Information Technology Co., Ltd. (PCITC) — Beijing, China
Back-end Engineering Intern — May. 2026 – Sep. 2026

*Contributed to an overseas service-station **Back Office System (BOS)** covering **6 core workflows** including procurement, goods receipt, inventory transfers, returns, stocktaking, and shift operations in a **Spring Boot**, **MyBatis-Plus**, **MySQL**, and **Docker Compose** environment.*

- Handled **20+ backend tickets** involving API behavior and data inconsistencies, tracing requests across **Controller–Service–Mapper** layers and using **Swagger/OpenAPI, SQL queries, application logs, and JDWP debugging** to identify root causes and validate fixes.
- Implemented and maintained **REST APIs** for inventory ops, including querying receipt records, updating transaction status, retrieving stock data, and persisting business entities through Spring Boot, MyBatis-Plus, and MySQL.
- Analyzed business logic for goods receipt confirmation and reversal, inventory ledger updates, stock transfers, **weighted-average costing**, and HOS synchronization to understand transactional data flow across the BOS.

INDIVIDUAL PROJECTS

JChatMind – AI Agent System
Full Stack / Backend Developer

Built a full-stack AI agent platform that combines multi-step reasoning, tool invocation, and RAG to support interactive knowledge-based conversations, using PostgreSQL/pgvector, SSE, and standardized REST APIs for model and tool integration.

- Built a full-stack AI agent system with a multi-step **Think-Execute loop**, tool invocation, and retrieval-augmented generation (RAG).
- Designed **modular agent orchestration** with dynamic model selection, parameter configuration, and extensible tool integration.
- Implemented **PostgreSQL/pgvector** retrieval and **Server-Sent Events (SSE)** for low-latency streaming responses.
- Integrated **DeepSeek and Ollama-hosted models** behind standardized REST APIs for flexible model switching.

Interactive Brainstem Learning Platform (Capstone)
Front End Developer

*Developed an interactive medical learning platform for students to explore brainstem anatomy through linked 3D/2D visualization, search, annotations, and quizzes, using **Next.js/React**, MongoDB, and Docker.*

- Collaborated in a four-person Agile team on a Honda-sponsored learning platform with an OSU College of Medicine faculty partner.
- Built with Next.js/React, MongoDB, and **Docker**; designed REST API schemas for terms, annotations, and quizzes.
- Implemented linked 3D-to-2D highlighting, search, pins, and a quiz mode with immediate feedback.

O-RAN edge-computing platform
Researcher

*Contributed to an experimental **O-RAN edge-computing platform** focused on containerized network functions, service communication, deployment automation, and **AI-assisted decision support.***

- Worked with **Docker** and **Kubernetes** to orchestrate containerized O-RAN network functions in the distributed research environment.
- Developed and debugged **REST APIs** for microservice communication and data exchange; integrated **AI agents** into the edge pipeline for experimental decision support.

PROFESSIONAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C#, SQL, HTML/CSS
Frameworks: Spring Boot, MyBatis, Node.js, FastAPI, Flask, Next.js, React, Tailwind
Tools: Docker, Kubernetes, Maven, Nginx, Redis, MySQL, MongoDB, Azure, Linux, Git